Omar Ayman Bakr

AI / LLM Engineer · Machine Learning

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LinkedIn GitHub Portfolio LeetCode · 500+ solved

SUMMARY

AI engineer who builds LLM systems and the infrastructure underneath them. Designed and shipped a retrieval-augmented document Q&A service now running in production: eleven model providers and two interchangeable databases behind one abstraction, streamed answers with page-level citations, and **455 automated tests** gating a pipeline that deploys itself to a live server. Trains models as well as serves them — a hierarchical LSTM reaching **85.6% accuracy** on group activity recognition, above the **81.9%** CVPR 2016 result on the same dataset. Mechatronics Engineering graduate; **500+ LeetCode problems** solved.

TECHNICAL SKILLS

- **LLM & Retrieval:** RAG, embeddings, vector search (pgvector, Qdrant), semantic chunking, token streaming (SSE), prompt design, LangChain, Ollama, OpenAI / Anthropic / Google / Cohere / NVIDIA NIM APIs
- **Backend:** FastAPI, Pydantic, async I/O, SQLAlchemy 2.0, Alembic, REST APIs, pytest
- **Machine Learning:** PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, NumPy, Pandas, Matplotlib, Seaborn
- **Deep Learning:** CNNs (ResNet), LSTMs, Transfer Learning, Focal Loss, Multi-GPU Training, Hydra, TensorBoard
- **Data & Infrastructure:** PostgreSQL, MongoDB, Docker Compose, GitHub Actions (CI/CD), Prometheus, Grafana, Linux, Git, uv
- **Programming:** Python, C++, SQL, C, MATLAB

PROJECTS

NotebookLLM-minus — Retrieval-Augmented Document Q&A Service | Code | *LLM / RAG · 2026–Present*

- Built a full RAG service in which **nothing above the factory layer names a vendor**: **eleven model providers** (six chat, five embedding — Anthropic, OpenAI, Google, Cohere, NVIDIA NIM, local Ollama) and **two interchangeable document stores** (MongoDB + Qdrant, or PostgreSQL + pgvector behind eight repository interfaces and Alembic migrations), each selected by configuration or per notebook.
- Answers **stream token by token over SSE** with model reasoning separated from the reply, and cite the source **page** rather than a chunk index — the exact passage highlighted from word coordinates extracted across a process pool, which cut a 274-page document from 520s to 79s.
- Made the model picker trustworthy by **probing rather than assuming**: capability and entitlement checks resolve what each host can actually do, trimming a vendor catalogue of 82 models to the 13 that answer a real request, so a model offered in the UI cannot fail on selection.
- **455 tests** gate a GitHub Actions pipeline that deploys over SSH to a live EC2 host (systemd + Docker Compose); Prometheus and Grafana track ingest duration, embedding latency and time-to-first-token.

Keywords: RAG, Vector Search, pgvector, Server-Sent Events, Repository Pattern, Alembic, CI/CD, Observability, Docker Compose

Volleyball Group Activity Recognition | Code | *Computer Vision · Deep Learning · 2025–2026*

- Built **eight PyTorch baselines** on the Volleyball Dataset (55 videos, 4,830 clips, 60+GB) — single-frame ResNet-50 through hierarchical player- and scene-level LSTMs — each isolating one architectural component and sharing one data loader, trainer and evaluator so the comparison measures the architecture and nothing else.
- Best model reaches **85.64% accuracy and 0.855 macro-F1**, above the CVPR 2016 published result of 81.9%. **Team-split pooling alone contributed +11.9 accuracy points and +0.154 macro-F1**, resolving the left/right confusion that limited every earlier baseline.
- Ran a **control experiment** to quantify what the hierarchy actually buys: a single YOLO26-x classifier on raw frames, using **none** of the player annotations, lands within ~7 points of the full hierarchical model.
- Trained with Hydra configs, AdamW and TensorBoard logging; multi-GPU via `nn.DataParallel` on dual T4s.

Keywords: Group Activity Recognition, Hierarchical LSTM, ResNet-50, YOLO, Ablation Study, Multi-GPU Training, Macro-F1

Variational Autoencoder on CelebA — Generative Modelling from Scratch | Code | *Generative AI · PyTorch · 2026*

- Implemented a **convolutional VAE from scratch in PyTorch** and trained it on CelebA, compressing $64 \times 64 \times 3$ faces into a **200-dimensional latent space** and generating new faces by sampling $z \sim \mathcal{N}(0, I)$ through the decoder alone, with no encoder at inference.
- Derived the **ELBO** — reconstruction term plus KL regulariser — and the **reparameterisation trick** that keeps gradients flowing through a stochastic bottleneck, publishing the full derivations as written notes beside the code: the foundation VQ-VAE and latent diffusion build on.

Keywords: Generative Models, VAE, ELBO, KL Divergence, Reparameterization Trick, Latent Space, CelebA, Convolutional Networks

[Credit Card Fraud Detection on Severely Imbalanced Data] Code
Classification · Model Benchmarking · 2025

- Benchmarked **11 models** on 285K transactions with a $< 0.2\%$ fraud rate, implementing **Focal Loss from its mathematical definition in PyTorch** to test whether an algorithmic fix beats resampling on extreme imbalance.
- **XGBoost won at F1 0.89 and ROC-AUC 0.98** — a 16% F1 gain over the logistic-regression baseline — while the focal-loss variants reached only 0.75–0.77, a negative result worth reporting: on this dataset gradient boosting beat the loss-function fix.

Keywords: Imbalanced Classification, Focal Loss (custom PyTorch), Gradient Boosting, Voting Ensemble, Average Precision, ROC-AUC

New York City Taxi Trip Duration Prediction Code
Regression · Feature Engineering · 2024

- Engineered **55 features from 10** by integrating five external sources (NYC DOT traffic, weather, borough geometry, holidays) and running a **local OSRM routing server in Docker** for real road distance and travel time on **1M+ trips**.
- Ridge Regression achieved **RMSE 0.397 and R^2 of 71.6%** on 224,734 held-out trips, with train and test scores within 0.003 RMSE of each other — the feature set generalised rather than memorised.

Keywords: Feature Engineering, OSRM / OpenStreetMap Routing, Geospatial Analysis, Ridge Regression, Polynomial Features

Tech Job Market Analysis with PostgreSQL Code
SQL · Data Analysis · 2024

- Analysed **700K+ tech job postings** across five countries in PostgreSQL — CTEs, window functions and multi-table joins over a star schema of company, posting and skill tables — then visualised the results in Python.
- Surfaced hiring and pay structure: data roles carry the largest senior-level salary premium, Python and SQL are near-universal requirements, and only ∼8% of postings offer remote work.

Keywords: Advanced SQL, CTEs, Window Functions, Star Schema, Aggregations, pgAdmin, Pandas, Data Visualization

ML Deep Dive — Core Algorithms Implemented from First Principles Code
Fundamentals · NumPy · 2025

- Implemented linear and polynomial regression, regularisation, logistic regression with log-loss, K-means, neural networks and ensemble methods **from scratch**, each with its mathematical derivation, complexity analysis, and a measured comparison of solution strategies — normal equation against gradient descent, iterative against vectorised.

Keywords: From-Scratch Implementations, Gradient Descent, Normal Equation, Vectorisation, Complexity Analysis, NumPy, Linear Algebra

EDUCATION

Bachelor of Science, Mechatronics Engineering 2019 – 2024
Helwan University, Cairo, Egypt — Graduated with Honors (Very Good, GPA 3.0/4.0)
Graduation Project: View Model Built a mathematical model in MATLAB for a green hydrogen generator, controlling input voltage and current while predictively monitoring safety parameters such as operating temperature.

LANGUAGES

Arabic — Native | **English** — Advanced